

Real-Time Smoking Detection Using YOLOv11: A Deep Learning-Based Surveillance Approach

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Abstract

Smoking continues to pose significant public health challenges, contributing to chronic diseases and environmental degradation. Enforcing smoking bans in public spaces demands efficient, automated surveillance systems. This project proposes a real-time smoking detection system leveraging YOLOv11, a state-of-the-art object detection model optimized for edge devices. The system aims to identify smoking behaviors and cigarettes in real-time, addressing limitations of existing methods, such as high computational costs and poor edge compatibility. The methodology encompasses dataset curation, model fine-tuning, optimization for edge deployment and rigorous evaluation using metrics like mean Average Precision (mAP), Precision, Recall and F1-score. Ethical considerations, including privacy protection and bias mitigation, are prioritized to ensure responsible deployment. This research contributes a novel YOLOv11-based solution for smoking detection, enhancing public health surveillance in resource-constrained environments.

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1 Introduction

Tobacco use remains a critical global public health issue, contributing to over 8 million deaths annually and causing severe health conditions such as lung cancer, cardiovascular diseases and respiratory disorders (World Health Organization, 2023). Beyond health impacts, cigarette waste significantly contributes to environmental pollution, with billions of cigarette butts discarded yearly, releasing toxic chemicals into ecosystems. Enforcing smoking bans in public spaces such as workplaces, transportation hubs, educational institutions and recreational areas is essential to mitigate these impacts but presents substantial challenges. Traditional monitoring methods, such as manual observation by security personnel, are labor-intensive, prone to human error and lack scalability for large or crowded areas.

Recent advancements in deep learning, particularly in object detection frameworks like the You Only Look Once (YOLO) series, offer a promising solution by enabling automated, accurate and scalable surveillance systems capable of identifying smoking behaviors in real time. YOLOv11, introduced in 2024, marks a significant leap in object detection technology with its lightweight architecture, incorporating advanced features like the C3K2 block, Spatial Pyramid Pooling Fast (SPPF) and C2PSA modules, which enhance efficiency and accuracy, especially for detecting small objects like cigarettes (Khanam & Hussain, 2024). Despite its potential, YOLOv11 has not been extensively explored for smoking detection as of mid-2025, presenting a unique opportunity to address the limitations of existing systems. This project proposes a YOLOv11-based real-time smoking detection system optimized for edge devices, such as IoT cameras and low-power hardware, to enable efficient surveillance in resource-constrained environments.

2 Research Problem Statement, Aim and Objectives

2.1 Research Problem Statement

Current smoking detection systems using deep learning models such as YOLOv5 or YOLOv8 achieve high accuracy but struggle with real-time performance and edge compatibility. Their high computational demands hinder deployment on resource-constrained devices like Raspberry Pi or Jetson Nano. Moreover, limited exploration of YOLOv11 in this domain creates an opportunity to design a lightweight, efficient detection system tailored for real-world applications in public health surveillance (Wang et al., 2023; Khanam & Hussain, 2024).

2.2 Research Aim

This study aims to develop a YOLOv11-based real-time smoking detection system optimized for edge devices, enabling accurate identification of smoking behaviors and cigarettes in public spaces. The system seeks to enhance public health surveillance by providing scalable, low-latency solutions that empower authorities to enforce smoking bans effectively.

2.3 Research Objectives

- Curate a comprehensive dataset of smoking-related images and videos, including diverse scenarios and environmental conditions.
- Fine-tune the YOLOv11 model to achieve high accuracy in detecting smoking behaviors and cigarettes.
- Optimize the model for edge deployment using techniques like quantization and pruning to ensure real-time performance.
- Evaluate the system's performance using quantitative metrics (e.g., mAP, Precision, Recall, F1-score) and qualitative feedback from usability testing.
- Address ethical considerations, including privacy protection and bias mitigation, to ensure responsible deployment.

2.4 Research Questions

- How accurate is the YOLOv11-based system in detecting smoking behaviors and cigarettes across diverse environmental conditions?
- Can the system achieve real-time performance on edge devices while maintaining high detection accuracy?
- How effectively does the system handle challenging scenarios, such as low-light conditions or small object detection?
- What are the ethical challenges in deploying a smoking detection system and how can they be mitigated?

4 Literature Review

Recent advancements in deep learning have transformed smoking detection through the use of convolutional neural networks (CNNs) and object detection frameworks, yet significant limitations remain in achieving real-time performance on edge devices. For instance, Wang et al. (2023) developed YOLOv8-MNC, a modified YOLO variant, achieving a high mean Average Precision (mAP) of 85.9% on a cigarette detection task using attention mechanisms like MHSA and advanced upsampling strategies. However, despite its strong accuracy, the model remains computationally intensive and is not optimized for real-time inference on low-power hardware.

In related work, Pundhir et al. (2021) applied region-based feature extraction techniques using YOLOv3 and achieved over 96% classification accuracy in detecting cigarette use, although their approach lacked efficiency benchmarks on edge platforms. These results indicate the viability of YOLO architectures for cigarette detection but highlight trade-offs between precision and deployability.

Lakatos et al. (2024) proposed a multimodal approach combining image and text data for smoking detection, achieving 74% accuracy. While promising, their method requires extensive processing resources, limiting its practical use in edge scenarios.

Small object detection, a key challenge in smoking detection, has been the focus of broader research as well. Cheng et al. (2022) provided a large-scale benchmark and analysis of small object detection tasks, emphasizing the importance of context preservation and feature resolution in object detection models. Similarly, Liu et al. (2021) conducted an extensive survey and performance evaluation of small object detection methods, noting that standard YOLO configurations struggle with objects under 32x32 pixels, a common case for cigarettes.

To overcome these limitations, new variations of YOLO have been proposed. Benjumea et al. (2021) introduced YOLO-Z, which enhances YOLOv5 with additional layers for small object localization and showed improved mAP in autonomous vehicle datasets. Ahn et al. (2024) presented enhancements to YOLOv5s including CBAM attention modules and focal-EIOU loss, yielding up to 10% mAP improvement on the VisDrone benchmark.

Finally, Khanam and Hussain (2024) provide an architectural overview of YOLOv11, highlighting its efficiency improvements such as the C3K2 block, Spatial Pyramid Pooling Fast (SPPF) and Cross-Stage Partial Self Attention (C2PSA) modules. These innovations are designed specifically to optimize inference speed while maintaining small-object accuracy, making YOLOv11 a promising candidate for edge-based surveillance systems.

This study addresses the underexplored potential of YOLOv11 in smoking detection, bridging the gap between high-accuracy detection and edge compatibility for public health surveillance.

5 Methodology

The development of the YOLOv11-based real-time smoking detection system is structured to span from August 20, 2025, to December 31, 2025, ensuring robust performance and compatibility with edge devices through a comprehensive and systematic approach. The methodology encompasses several critical phases, each designed to address specific technical and practical challenges in achieving accurate, efficient and scalable smoking detection.

5.1 Dataset Preparation

The initial phase focuses on creating a robust dataset to train and validate the YOLOv11 model. A comprehensive dataset is curated from public repositories such as CfgDet, Kaggle and Roboflow, ensuring a diverse collection of images capturing smoking behaviors and cigarettes across various scenarios. These scenarios include indoor and outdoor environments, different lighting conditions and diverse demographic groups to enhance model generalizability. Annotation of the dataset involves a combination of manual labeling and automated techniques to create precise bounding boxes for cigarettes and smoking actions. To ensure high-quality annotations, domain experts validate the labels, addressing potential inconsistencies and errors. Data augmentation techniques, including rotation, flipping, scaling and cropping, are applied to increase dataset variability and robustness against real-world conditions, such as varying angles, distances and occlusions. This approach ensures the dataset is representative of the diverse contexts in which the system will operate, minimizing biases and improving detection accuracy.

5.2 Model Fine-Tuning

The YOLOv11 nano variant, with approximately 15 million parameters, is selected as the core model due to its balance of efficiency and performance, making it suitable for edge deployment. Fine-tuning is performed using transfer learning, leveraging pre-trained weights provided by Ultralytics to initialize the model. This process optimizes key components of YOLOv11, such as the C3K2 block for efficient feature extraction and the Spatial Pyramid Pooling Fast (SPPF) module for improved small object detection, specifically tailored for identifying cigarettes. Training is conducted on a high-performance GPU, such as an AWS EC2 instance, with a batch size of 8 to ensure computational efficiency. The finetuning process involves adjusting hyperparameters, such as learning rate and momentum, to optimize convergence and performance on the smoking detection task. Validation is performed iteratively to monitor model performance and prevent overfitting, ensuring the model generalizes well to unseen data.

5.3 Model Optimization for Edge Deployment

To enable deployment on resource-constrained edge devices, such as Raspberry Pi 4 or NVIDIA Jetson Nano, the fine-tuned YOLOv11 model undergoes optimization. Techniques such as quantization to 8-bit precision and model pruning are applied to reduce computational complexity and memory footprint while maintaining detection accuracy. Quantization minimizes the model's size by converting floating-point weights to integers, enabling faster inference on low-power hardware. Pruning removes redundant or less significant network connections, further streamlining the model. The optimized model is exported to the ONNX format to ensure compatibility with edge devices and to leverage OpenCV for efficient input/output operations. Testing is conducted on target edge devices to verify real-time performance, ensuring low-latency inference suitable for surveillance applications.

5.4 Evaluation

The system's performance is rigorously evaluated using standard object detection metrics to ensure its reliability and effectiveness in detecting smoking behaviors and cigarettes. The primary metrics used include Precision, Recall, F1-score

Precision measures the proportion of correctly predicted positive detections among all positive detections made by the model:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP is the number of true positives and FP is the number of false positives.

Recall quantifies the proportion of actual positives correctly detected by the model:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN is the number of false negatives.

The **F1-score** is the harmonic mean of precision and recall, providing a balanced measure that captures the overall effectiveness of the model:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

To ensure robustness, evaluation is performed on diverse test sets representing various environments and conditions. Quantitative assessments with these metrics are complemented by qualitative evaluations involving visual inspection of detection results, confirming the model's practical usability in real-world scenarios.

5.5 System Architecture

The system architecture for the YOLOv11-based smoking detection system is designed to ensure real-time performance, edge compatibility and ethical compliance, aligning with the project's goal of scalable public health surveillance. The architecture integrates multiple components to process video inputs, detect smoking behaviors and generate actionable outputs while prioritizing privacy and efficiency. Below is the system architecture diagram in Mermaid syntax, followed by a detailed explanation of its components.

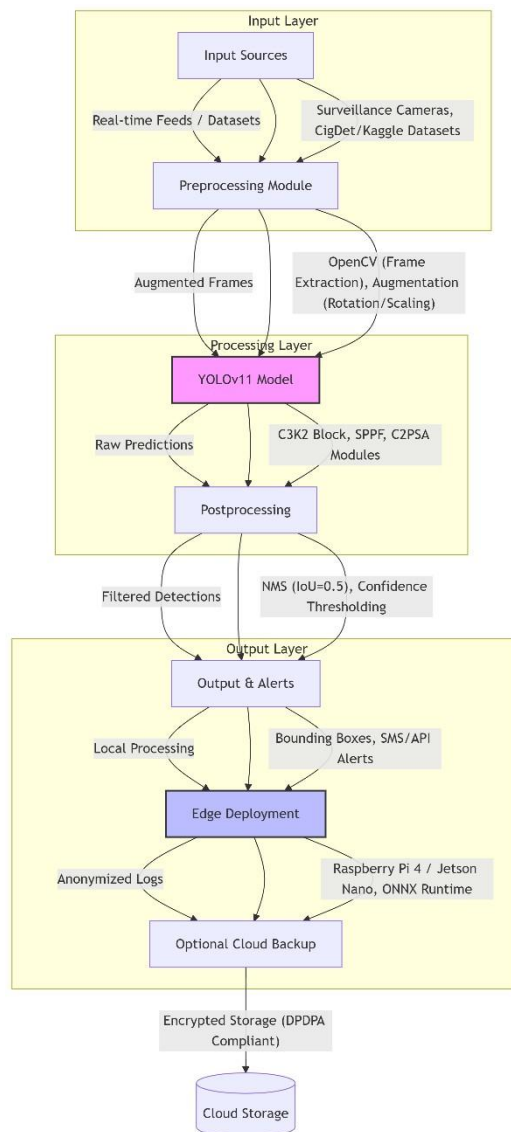


Figure 1 System Architecture

5.5.1 Key Components Explained

Input Sources: The system processes real-time video feeds from surveillance cameras or prerecorded videos, supplemented by public datasets like CigDet and Kaggle for training and validation, ensuring a robust foundation for model development.

Preprocessing Module: Video frames are extracted using OpenCV for real-time processing. Data augmentation techniques, including rotation ($\pm 30^\circ$), scaling ($0.8\text{--}1.2\times$) and flipping ($p=0.5$), enhance dataset diversity. Annotations, such as bounding boxes for cigarettes and smoking actions, are created and validated using Roboflow to ensure high-quality training data.

YOLOv11 Model: The core of the system, the YOLOv11 nano/small variant, leverages the C3K2 block for efficient feature extraction, Spatial Pyramid Pooling Fast (SPPF) for improved small object detection (e.g., cigarettes) and C2PSA for spatial attention to

focus on relevant regions. The model uses a loss function balancing localization and classification errors to optimize performance.

Postprocessing: Non-Maximum Suppression (NMS) eliminates duplicate detections, while thresholding filters predictions with confidence scores below 0.5, ensuring reliable outputs.

Output s Alerts: The system overlays bounding boxes on video streams to visualize detections in real time. Alerts for smoking violations are sent via SMS or API and anonymized logs ensure compliance with privacy regulations like India's Digital Personal Data Protection Act (DPDPA), 2023.

Edge Deployment: Optimized for low-power devices like Raspberry Pi 4 or NVIDIA Jetson Nano, the system uses ONNX for lightweight inference and OpenCV for input/output operations, enabling real-time performance in resource-constrained environments.

Optional Cloud Backup: Anonymized logs are stored for audit purposes or model re-training, maintaining ethical compliance and supporting system scalability.

Architecture Highlights: The edge-first design uses quantized (8-bit) and pruned YOLOv11 variants for low-latency inference. Ethical safeguards, such as data anonymization and diverse datasets for bias mitigation, ensure responsible deployment. The modular design facilitates integration with existing surveillance systems, enhancing scalability.

This architecture ensures a robust, efficient and ethically compliant system for real-time smoking detection, tailored for public health surveillance.

5.6 Project Flowchart

To illustrate the logical progression of the research process, a flowchart is provided below, capturing the key phases of the project as outlined in the report. The flowchart is presented in Mermaid syntax, followed by an explanation of each step.

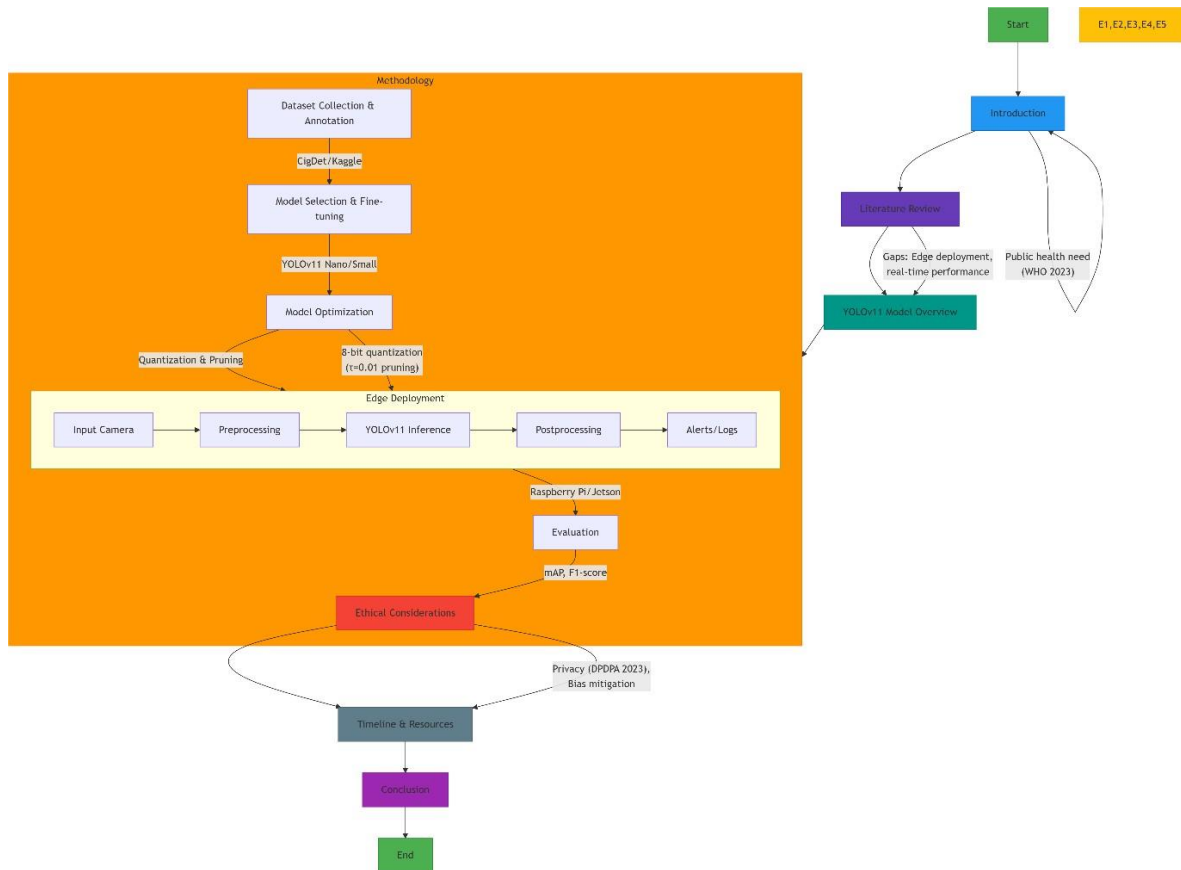


Figure 2 Flowchart

6 Ethical and Practical Considerations

Throughout the methodology, ethical considerations are integrated to ensure responsible development and deployment. Privacy is prioritized by processing data locally on edge devices to minimize data transmission and storage. Anonymized logs are maintained for audit purposes, complying with regulations such as India's Digital Personal Data Protection Act (DPDPA), 2023. Bias mitigation is addressed by ensuring the dataset includes diverse demographics and environments, reducing the risk of skewed detection outcomes. Regular audits by computer vision experts are planned to maintain fairness and reliability, ensuring the system aligns with ethical standards for public health surveillance.

This methodology provides a structured and detailed framework for developing a YOLOv11based smoking detection system, balancing technical rigor with practical and ethical considerations to achieve a scalable, efficient and responsible solution.

7 Timeline and Resource

7.1 Timeline

Phase	Duration	Tasks	Deliverables
Dataset Preparation	Aug 20 – Sep 15, 2025	Curate and annotate dataset; apply augmentation.	Curated smoking detection dataset.
Model Fine-Tuning	Sep 16 – Oct 10, 2025	Fine-tune YOLOv11; validate with experts.	Fine-tuned YOLOv11 model.
Model Optimization	Oct 11 – Nov 5, 2025	Apply quantization and pruning; test performance.	Optimized model for edge deployment.
Edge Deployment	Nov 6 – Nov 30, 2025	Deploy on Raspberry Pi/Jetson Nano; integrate OpenCV/ONNX.	Deployed system.
Evaluation	Dec 1 – Dec 31, 2025	Conduct quantitative and qualitative evaluations.	Evaluation report.

7.2 Resource Requirements

Category	Resources	Purpose
Hardware	Raspberry Pi 4, NVIDIA Jetson Nano, AWS EC2 (g4dn.xlarge)	Model training and deployment.
Software	Python, OpenCV, ONNX, Ultralytics YOLOv11	Development and inference.
Data	CigDet, Kaggle, Roboflow datasets	Training and evaluation

8 Conclusion

This project proposes a YOLOv11-based real-time smoking detection system to address the need for efficient, scalable surveillance in public spaces. By leveraging YOLOv11's lightweight architecture, the system ensures edge compatibility while maintaining high accuracy. The methodology includes dataset curation, model optimization and rigorous evaluation, with ethical considerations ensuring responsible deployment. The system's scalability and adaptability make it a significant contribution to public health surveillance, with potential applications in other detection tasks.

9 References

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